

Kaibo Tang

☎ 631-235-5125 ✉ tangkaibo@gmail.com 🌐 github.com/kvttt 💼 linkedin.com/in/kaibotang

EDUCATION

Boston University

Boston, MA

Ph.D. in Electrical Engineering

September 2025 – Present

- **GPA (cumulative):** 4.0/4.0
- **Selected ECE Courses:** Digital Signal Processing, Optimization Theory and Methods, Digital Image Processing and Communication, Image Reconstruction and Restoration, Stochastic Processes
- **Selected BME Courses:** Introduction to Medical Imaging
- **Honors and Awards:**
 - Distinguished Electrical Engineering Fellowship
 - College of Engineering Convergent Fellowship

University of North Carolina at Chapel Hill

Chapel Hill, NC

B.S.P.H. in Biostatistics, B.S. in Mathematics, and minor in Chemistry

August 2021 – May 2025

- **GPA (cumulative):** 3.929/4.0
- **Selected Biostatistics Courses:** Principles of Epidemiology, Principles of Experimental Analysis, Probability and Statistical Inference, Sample Survey Methodology, Design of Public Health Studies
- **Selected Mathematics Courses:** Differential Equations, Linear Algebra, Advanced Calculus, Introductory Analysis, Introductory Functional Analysis
- **Pre-Med Courses:** Molecular Biology and Genetics, Cell and Developmental Biology, Anatomy and Physiology (Lab), General Chemistry, Analytical Chemistry (Lab), Organic Chemistry (Lab), Biological Chemistry
- **Additional Electives:** Image Processing and Analysis
- **Honors and Awards:**
 - Phi Beta Kappa, 2023
 - Honors Carolina
 - Summer Undergraduate Research Fellowship, 2022
 - Accelerated Research Program

SELECTED PUBLICATIONS AND MANUSCRIPTS

- **Kaibo Tang**, Zijing Dong, Fuyixue Wang. Accelerated Romer-EPTI with Joint Spatial-q-Space Attention Network-Constrained Reconstruction for Rapid Mesoscale Diffusion MRI. *Under review at Magnetic Resonance in Medicine*.
- **Kaibo Tang**, Zijing Dong, Lawrence L. Wald, Fuyixue Wang. Rapid multi-shell mesoscale dMRI using accelerated Romer-EPTI with joint x-q attention network constrained reconstruction. *ISMRM 2026 (oral)*.
- **Kaibo Tang**, Xiuyu Dong, Zhengwang Wu, Laifa Ma, Sheng-Che Hung, He Zhang, Weili Lin, Gang Li. Surface-Guided Construction of 4D Volumetric Atlases of Fetal Brains. *MICCAI 2025 (MLMI workshop)*.
- **Kaibo Tang**. A Selection of Distributions and Their Fourier Transforms with Applications in Magnetic Resonance Imaging. *arXiv*. 2025.
- **Kaibo Tang**, Zijing Dong, Lawrence L. Wald, Fuyixue Wang. Accelerated Romer-EPTI using physics-driven joint x-q attention-network regularized reconstruction for fast mesoscale diffusion MRI. *ISMRM 2025 (power pitch)*.
- **Kaibo Tang**, Liangjun Chen, Zhengwang Wu, Fenqiang Zhao, Ya Wang, Weili Lin, Li Wang, Gang Li. Generation of Anatomy-Realistic 4D Infant Brain Atlases with Tissue Maps Using Generative Adversarial Networks. *ISBI 2024 (oral)*.

RESEARCH EXPERIENCE

Advanced MRI Acquisition Lab

March 2024 – Present

Advisors: Drs. Fuyixue Wang and Zijing Dong

A. A. Martinos Center for Biomedical Imaging, Massachusetts General Hospital, Harvard Medical School

- Developed a novel reconstruction with plug-and-play prior to achieve up to six-fold accelerated Romer-EPTI for rapid multi-shell mesoscale dMRI in vivo. This work has resulted in two abstracts at ISMRM 2025 (power pitch) and ISMRM 2026 (oral), and a manuscript submitted to MRM.

UNC BRAIN Lab

September 2021 – May 2025

Advisor: Dr. Gang Li

Biomedical Research Imaging Center (BRIC), UNC Chapel Hill

- Conducted preliminary study investigating the effect of auxiliary data on improving learning-based image registration for infant brain MRI. This work is supported by the Summer Undergraduate Research Fellowship (SURF) award (\$4,000) from the UNC Office of Undergraduate Research.
- Developed a learning-based framework for constructing 4D infant brain volumetric atlases. This work has resulted in a 4-page paper at ISBI 2024 (oral).
- Further developed the previous framework to incorporate anatomical information from reconstructed pial and white matter surfaces to model fetal brain developmental trajectory. This work has resulted in a 8-page paper at the MLMI workshop of MICCAI 2025.

Open-Source Software Development

November 2023 – Present

- Made major contributions to MONAI, an open-source deep learning framework for medical imaging maintained by NVIDIA. These contributions include a re-implementation of VoxelMorph, one of the most popular image registration frameworks, and Diffusion Loss, a popular regularizer for image registration.
- Contributed to MONAI Tutorials, including one tutorial on image registration with VoxelMorph.
- Contributed to DIPY, a library for the analysis of diffusion MRI data. The contribution includes a function that enables interpolation of tensor-valued spherical functions.

ACADEMIC SERVICE

- **Reviewer:** MedIA, IEEE JBHI, MICCAI 2023, ISBI 2025, ISBI 2026, MICCAI 2026
- **Lab Assistant:** Fundamentals of Human Anatomy and Physiology Laboratory (UNC Chapel Hill, Spring 2023, Fall 2023, Spring 2024, Fall 2024)

VOLUNTEERING

- **Undergraduate Volunteer:** UNC Hospital Emergency Department (January 2023 – December 2024)

SKILLS

- **Languages:** Python, $\text{\LaTeX}$, R, HTML, Markdown
- **Professional Softwares:** MATLAB, SAS, FreeSurfer, FSL, ANTs, ITK, VTK
- **Others:** Git, Docker, PyTorch, TensorFlow, DIPY, MONAI

SELECTED PROJECTS, VIDEOS, AND NOTES

- **Selected projects:** See kvttt.github.io/projects, also see github.com/kvttt.
- **Videos produced:** See kvttt.github.io/videos.
- **Notes:** See kvttt.github.io/notes.